

HW1.4

RJ Cass

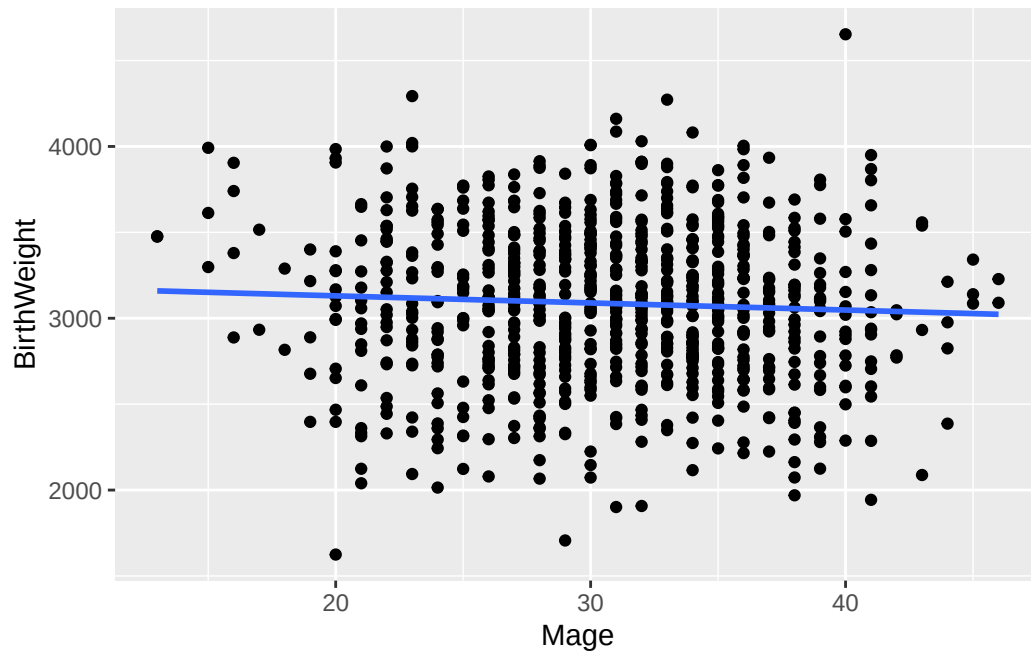
```
suppressPackageStartupMessages(library('ggplot2'))
suppressPackageStartupMessages(library('dplyr'))
suppressPackageStartupMessages(library('car'))
suppressPackageStartupMessages(library('lmtest'))
suppressPackageStartupMessages(library('multcomp'))
data <- read.table("BirthWeights.txt", sep = " ", header = TRUE, as.is = FALSE)
```

HW 1.1

1.

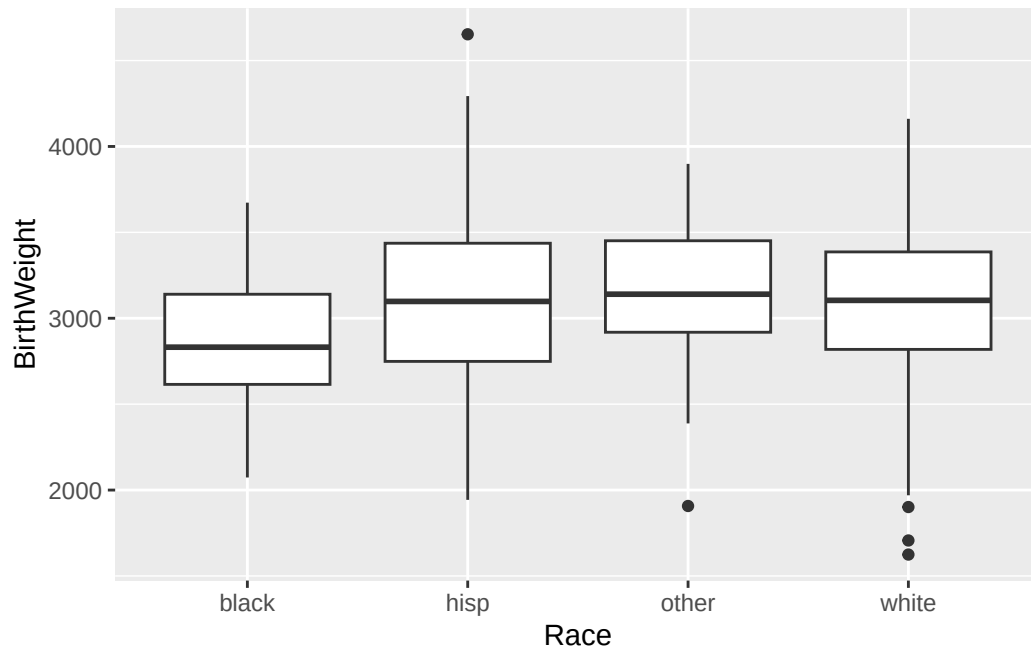
```
ggplot(data, aes(x=Mage, y=BirthWeight)) +
  geom_point() +
  geom_smooth(method="lm", se=FALSE)
```

`geom\_smooth()` using formula = 'y ~ x'



2.

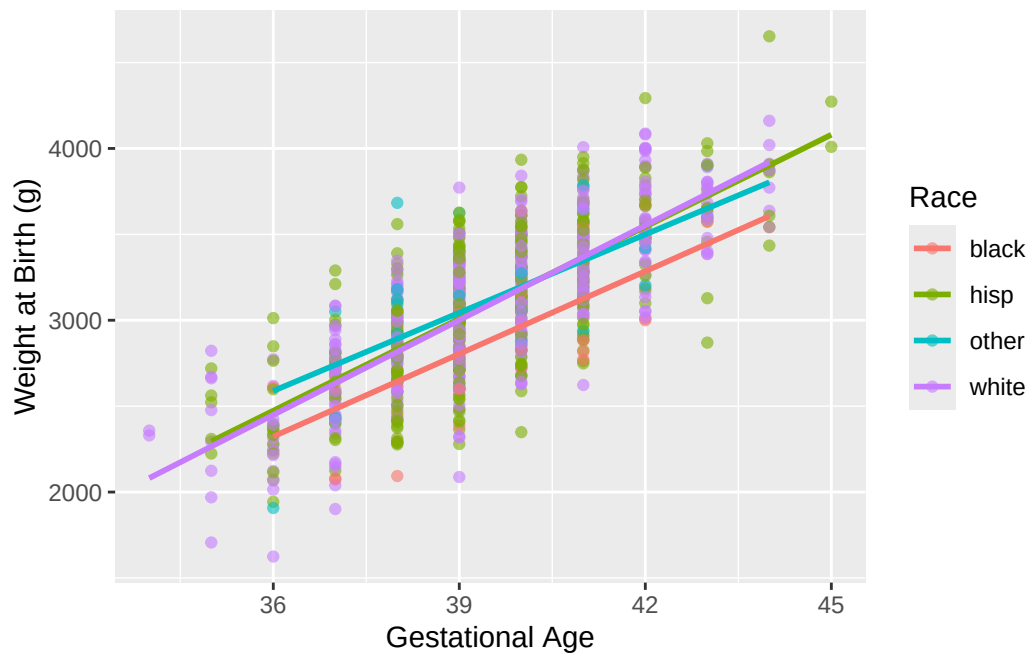
```
ggplot(data, aes(x=Race, y=BirthWeight)) +  
  geom_boxplot()
```



3.

```
ggplot(data, aes(x = Gage, y = BirthWeight, color = Race)) +  
  geom_point(alpha = 0.6) +  
  geom_smooth(method = "lm", se = FALSE) +  
  labs(x = 'Gestational Age', y = "Weight at Birth (g)")
```

`geom\_smooth()` using formula = 'y ~ x'



4.

```
data %>%
  group_by(Gen) %>%
  summarise(correlation = cor(BirthWeight, Mage))
```

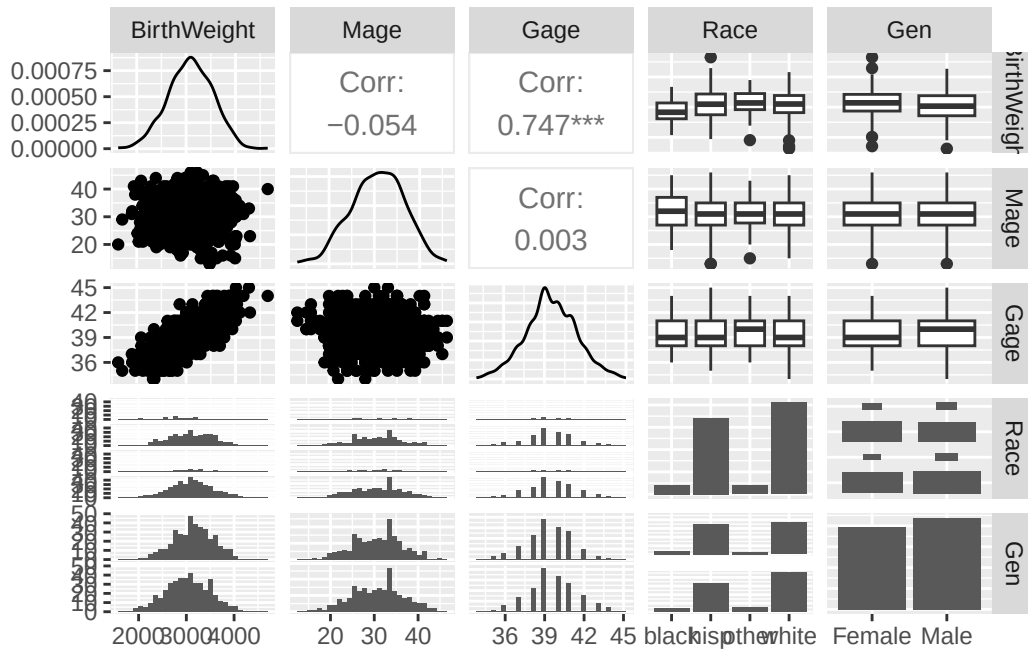
```
# A tibble: 2 x 2
  Gen      correlation
  <fct>      <dbl>
1 Female -0.0261
2 Male   -0.0745
```

5.

```
GGally::ggpairs(data)
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
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```



HW 2.2

1.

```
# Design matrix
X <- model.matrix(~ Mage + Gage + Race + Gen, data = data)
y <- matrix(data[, 1], ncol = 1)

# beta = (X'X)^(-1)*X'y
beta <- solve(t(X) %*% X) %*% t(X) %*% y

# S2
n <- nrow(X)
p <- ncol(X)
resids <- y - (X %*% beta)
S2 <- sum(resids^2) / (n-p)
```

```
lm <- lm(BirthWeight ~ ., data = data)
```

```
beta
```

```

              [,1]
(Intercept) -4120.542409
Mage         -3.793751
Gage         182.742497
Racehisp     198.747954
Raceother    241.582827
Racewhite    204.888197
GenMale      -169.348562

```

```
lm$coefficients
```

```

(Intercept)      Mage      Gage      Racehisp      Raceother      Racewhite
-4120.542409    -3.793751    182.742497    198.747954    241.582827    204.888197
      GenMale
    -169.348562

```

```
S2
```

```
[1] 79277.09
```

```
summary(lm)$sigma^2
```

```
[1] 79277.09
```

2.

```

# Manual vs. LM
table(round(X %*% beta, 3) == round(lm$fitted.values, 3))

```

```

TRUE
832

```

3.

```
# Manual vs. LM  
table(round(y - X%%beta, 3) == round(lm$residuals, 3))
```

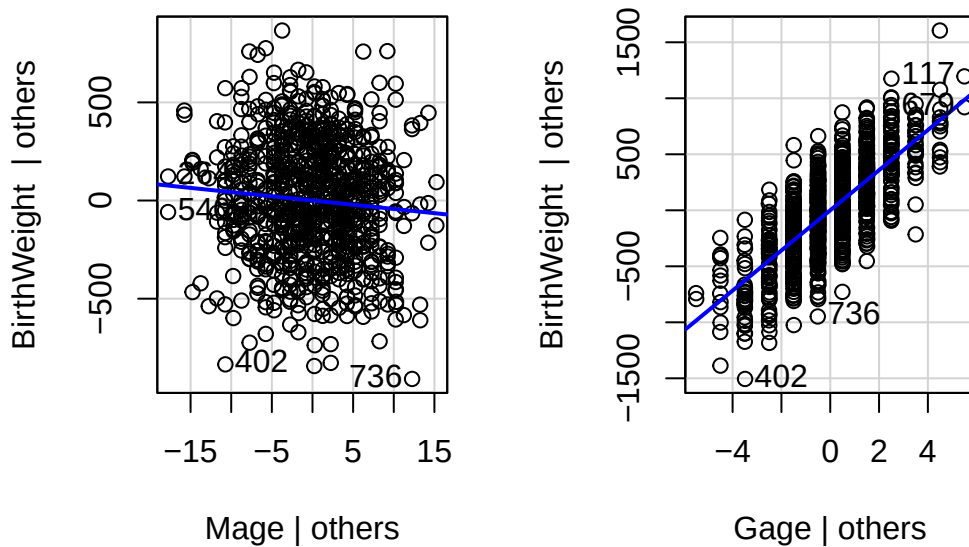
TRUE
832

HW 1.3

1.

```
mod <- lm(BirthWeight ~ Mage + Gage, data = data)  
avPlots(mod)
```

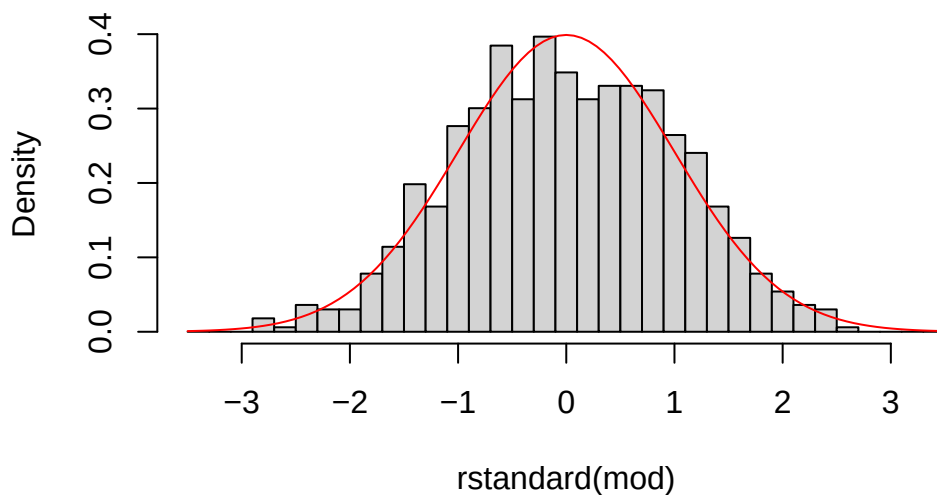
Added-Variable Plots



2.

```
mod <- lm(BirthWeight ~ ., data = data)
hist(rstandard(mod), freq = FALSE, breaks = seq(from = -3.5, to = 3.5, by = .2))
curve(dnorm(x), add = TRUE, col = 'red')
```

Histogram of rstandard(mod)



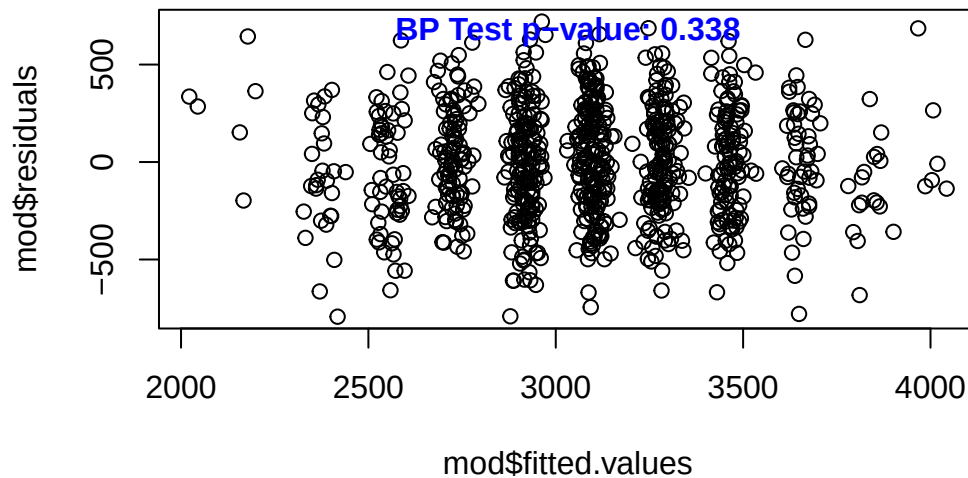
```
ks.test(rstandard(mod), "pnorm")
```

Asymptotic one-sample Kolmogorov-Smirnov test

```
data:  rstandard(mod)
D = 0.028947, p-value = 0.4884
alternative hypothesis: two-sided
```

3.

```
bp <- bptest(mod)
p_val <- bp$p.value
p_label <- paste("BP Test p-value:", format.pval(p_val, digits = 3))
plot(mod$fitted.values, mod$residuals)
mtext(p_label, side = 3, line = -1, font = 2, col = 'blue')
```



4.

```
summary(mod)$r.squared
```

```
[1] 0.6064689
```

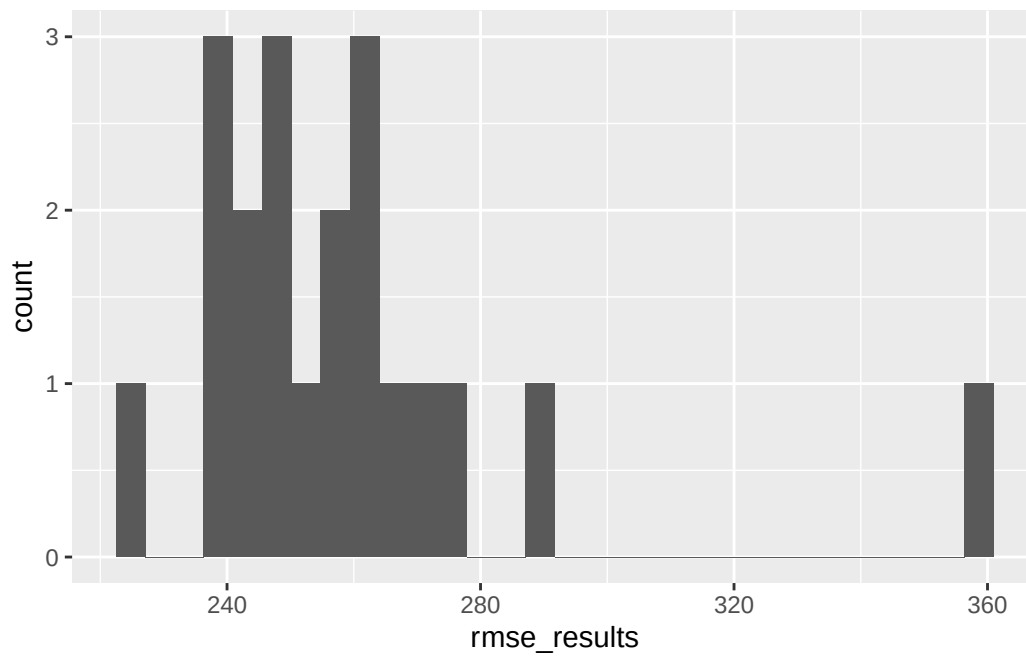
5.

```
cross_validate <- function(fold_num){
  cv_data <- data[folds == fold_num, ]
  cv_mod <- lm(BirthWeight ~ ., data = cv_data)
  rmse <- sqrt(mean(cv_mod$residuals^2))
  rmse
}

set.seed(1337)
K <- 20
folds <- rep(1:K, length = nrow(data)) %>% #Rep K folds n times
  sample()
## Using sapply()
rmse_results <- sapply(1:K, FUN=cross_validate)

## Plot the Results
ggplot() + geom_histogram(mapping=aes(x=rmse_results))
```

``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



HW 1.4

1

$$SE(\beta_{Gage}) = s\sqrt{a'(X'X)^{-1}a}$$

```
summary(mod)
```

Call:

```
lm(formula = BirthWeight ~ ., data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-793.32	-196.79	-5.24	208.89	720.63

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-4120.542	218.050	-18.897	< 2e-16 ***

Mage	-3.794	1.680	-2.259	0.024171	*
Gage	182.742	5.256	34.770	< 2e-16	***
Racehisp	198.748	46.617	4.263	2.25e-05	***
Raceother	241.583	62.639	3.857	0.000124	***
Racewhite	204.888	46.177	4.437	1.04e-05	***
GenMale	-169.349	19.677	-8.607	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 281.6 on 825 degrees of freedom

Multiple R-squared: 0.6065, Adjusted R-squared: 0.6036

F-statistic: 211.9 on 6 and 825 DF, p-value: < 2.2e-16

```
confint(mod)
```

	2.5 %	97.5 %
(Intercept)	-4548.540061	-3692.5447576
Mage	-7.090783	-0.4967197
Gage	172.426278	193.0587163
Racehisp	107.246752	290.2491562
Raceother	118.631931	364.5337216
Racewhite	114.250896	295.5254974
GenMale	-207.970846	-130.7262789

```
a <- t(matrix(c(0,0,1,0,0,0,0)))
s <- 281.6 # Residual Standard Error from summary
se_gage <- as.numeric(s * sqrt(a%*%solve(t(X)%*%X)%*%t(a)))
est_gage <- mod$coefficients[3]
gage_int <- est_gage + c(-1, 1)*qt(.975, df=nrow(data)-ncol(X))*se_gage

gage_int
```

```
[1] 172.4249 193.0601
```

2

Q = number of variables being tested

$$F = \frac{(R_{All}^2 - R_{Red}^2)/Q}{(1 - R_{All}^2)/(n - P - 1)}$$

```

Q <- 6
R2_all <- .6065
R2_red <- 0

F_stat_manual <- ((R2_all - R2_red)/Q)/((1-R2_all)/(nrow(X) - 7-1))
F_stat_R <- summary(mod)$fstatistic[1]

```

Manual F-stat: 211.6713257 R F-stat: 211.900591

The difference is just because I rounded the R2 value in the manual case.

3

```

int_mod <- lm(BirthWeight ~ . + Gage:Gen, data = data)
anova(mod, int_mod)

```

Analysis of Variance Table

```

Model 1: BirthWeight ~ Mage + Gage + Race + Gen
Model 2: BirthWeight ~ Mage + Gage + Race + Gen + Gage:Gen
  Res.Df    RSS Df Sum of Sq    F Pr(>F)
1     825 65403597
2     824 65369557  1      34040 0.4291 0.5126

```

Large p-value, so we accept the null-hypothesis that there is no difference between the models, meaning the interaction is not significant

4

```

hyp_diff <- glht(mod, linfct="5*Mage - Racehisp + GenMale == 0")
confint(hyp_diff)

```

Simultaneous Confidence Intervals

```
Fit: lm(formula = BirthWeight ~ ., data = data)
```

```
Quantile = 1.9628
95% family-wise confidence level
```

Linear Hypotheses:

	Estimate	lwr	upr
5 * Mage - Racehisp + GenMale == 0	-387.0653	-486.1285	-288.0021

5

```
new_data <- data.frame(Mage = 25, Gage = 30, Race = "white", Gen = "Female")
predict(mod, newdata = new_data, interval = "prediction", level = 0.75)
```

	fit	lwr	upr
1	1471.777	1141.965	1801.589